

Lithic Debitage and Unmod. Stone

Monticello Field School | 2026

Last updated by Iris Puryear, June 3, 2026 (original by CAS)

Lithics

Overview

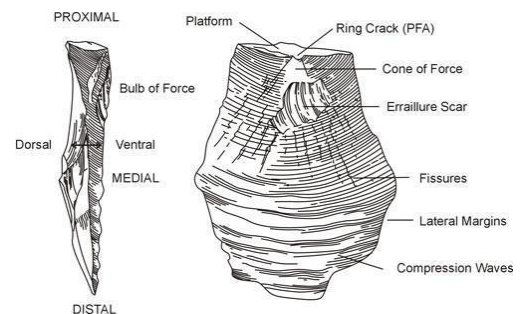
- Stone tool manufacturing via flint knapping or grinding to create:
 - Projectile points
 - Celts, chisels, adzes, etc.
 - Also can't forget other stone items like pipes, vessels, gunflints, etc.
- We will be focusing on lithic debitage (waste material) created by flint-knapping processes.
- Important indicator of Native American people (likely ancestral Monacans at Monticello) producing and retouching projectile points and flaked stone tools.

Quartz

- Many naturally occurring veins and impurities
- Not ideal for flint knapping compared to other materials
- But it's here, and they're using it.
- **Majority of lithics we find at Monticello are quartz/quartzite.**

Debitage

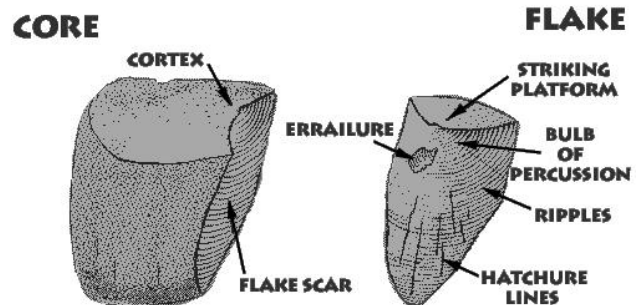
- Expect to see two biproducts of flintknapping: flake and shatter
- **Question: How do we tell the difference between natural and cultural breakage?**
- Worked with geologists, lithic experts, and utilized experimental archaeology to create a set of rules.



Flake

Pieces of debitage that have been intentionally removed or fractured from the core material. This process creates conchoidal fractures in the stone. Conchoidal fractures are characterized by specific attributes that we use to distinguish flakes from unintentional debris or shatter.

- Ventral surface is smooth (interior side of the flake that has broken away from the stone—the fresh break); opposite side (dorsal surface) may or may not have cortex and/or previous flake scars
- Striking platform: point of contact where force was applied to remove the flake (at the “top” of the flake or what is referred to as the “proximal end”)



- Feathering: slight curve or “feathering” of the profile towards the distal end (the end opposite the striking platform)
- Bulb of percussion: raised hump or “bulb” directly beneath the striking platform on the ventral surface, forms on impact
- Eraillure scar: small scar or removal that sometimes appears on the bulb of percussion, caused by competing waves of force traveling through stone from multiple impacts
- Can also sometimes see “ripples” or striations on ventral surface from impact (waves of force traveling through material)

Note: Not all flakes will exhibit every one of these attributes, and some will be faint or ambiguous. This is especially true when a flake has been broken. This is part of what makes lithic identification so difficult.

Core

- i.e., the parent material, the object being worked (various subtypes)
- Must see:
 - **Flake scar**: slightly concave, something clearly popped off.
- Also can see: bifacial cores (occupy a liminal area—may be used as tools themselves or could have been discarded on their way to become a tool)

Shatter

- The unintentional secondary debris resulting from the flintknapping process.
- Expect it to be obviously popped off something else and be blocky.
- Again, difficult with quartz - and how do we distinguish from natural processes?
- These will be the fragments not directly hit by the hammerstone, but that were adjacent to at least two of the following: 1) core; 2) flake; 3) other shatter.
- Therefore, must see the following:
 - **Exhibit at least two planes of clear breaks**
 - **The two planes must be touching.**
 - There will never be a situation in which the only two culturally caused breaks would be present on opposite sides of a piece of shatter.

Unmodified stone

Natural stone types we expect to see at on the mountaintop:

- Quartz, again.
- Quartzite and quartz sandstone
- Greenstone
 - Decomposing greenstone (typically crumbly, yellowish/orangey in color)

Architectural stone:

Architectural stone from the historic period will often exhibit attributes consistent with lithic production. We can differentiate these architectural debris from those associated with flint knapping by material type, because we know that certain stone materials were brought to the mountaintop in the Jefferson period specifically for architectural purposes.

- Arkose (“Alaskite”)
 - A variety of quartzite. Visibly grainy, pinkish in color
- Slate and phyllite
 - Flat, laminated, fine grained, bluish grey to greenish in color
- Limestone
 - Typically greyish or bluish-grey, chalky, somewhat laminated

WHEN IT DOUBT, BRING IT INTO THE LAB!